

Image #1: Screenshot Capture - 2026-06-11 - 01-09-22.png

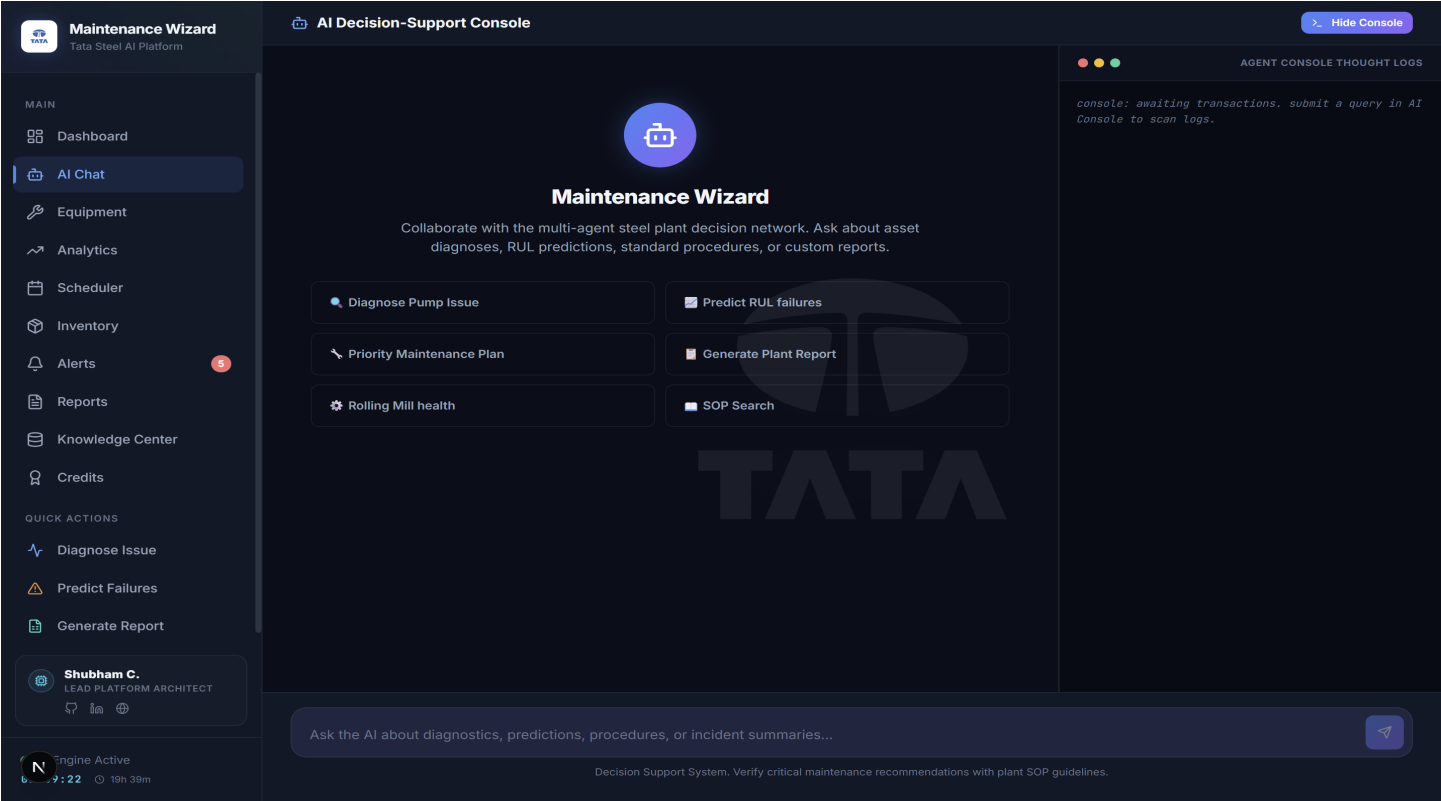


Image #2: Screenshot Capture - 2026-06-11 - 01-10-35.png



Image #3: Screenshot Capture - 2026-06-11 - 01-10-48.png



Image #4: Screenshot Capture - 2026-06-11 - 01-10-55.png

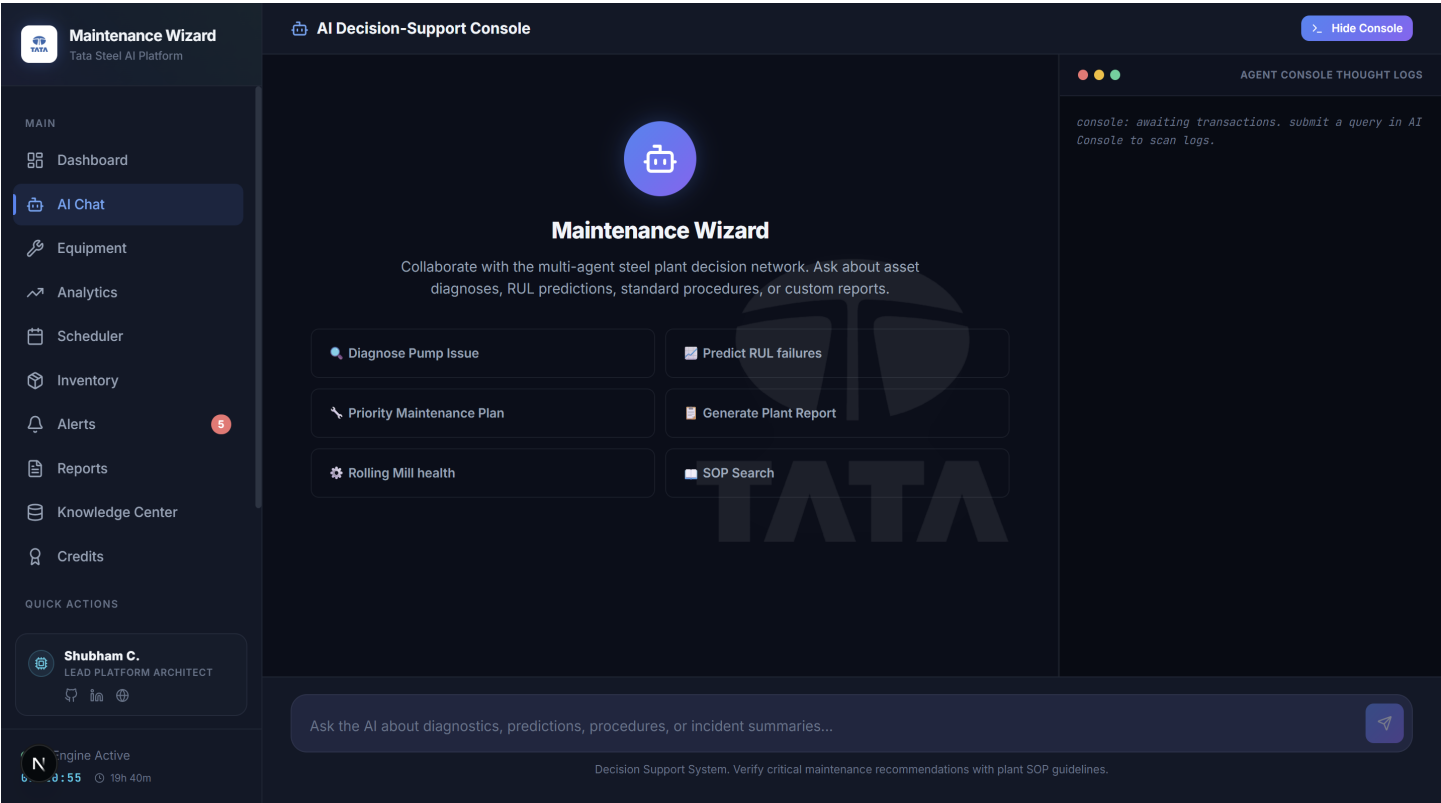


Image #5: Screenshot Capture - 2026-06-11 - 01-11-06.png

Maintenance Wizard
Tata Steel AI Platform

MAIN

Dashboard
AI Chat
Equipment
Analytics
Scheduler
Inventory
Alerts
Reports
Knowledge Center
Credits

QUICK ACTIONS

Shubham C.
LEAD PLATFORM ARCHITECT

Engine Active
11:05 19h 41m

Equipment Fleet

25 assets monitoring active across 6 sectors

All (25)

Blast Furnace (5)

Steel Melting Shop (5)

Rolling Mill (5)

Coke Oven (3)

Sinter Plant (3)

Power Plant (4)

Mill Gearbox #1
Rolling Mill • Helical Gearbox

Health Score
39.4%

Criticality: CRITICAL
Risk Level: CRITICAL
Asset ID: RM-GB-001

Steam Turbine Generator
Power Plant • Steam Turbine

Health Score
49.6%

Criticality: CRITICAL
Risk Level: HIGH
Asset ID: PP-TG-001

Blast Furnace Cooling Pump #1
Blast Furnace • Centrifugal Pump

Health Score
49.9%

Criticality: CRITICAL
Risk Level: HIGH
Asset ID: BF-CP-001

Sinter Fan Main Blower
Sinter Plant • Centrifugal Fan

Health Score
50.3%

Criticality: CRITICAL
Risk Level: HIGH
Asset ID: SP-FM-001

Continuous Caster #1
Steel Melting Shop • Continuous Casting Machine

Health Score
50.7%

Criticality: CRITICAL
Risk Level: HIGH
Asset ID: SMS-CC-001

Hot Blast Blower
Blast Furnace • Turbo Blower

Health Score
51.5%

Criticality: CRITICAL
Risk Level: HIGH
Asset ID: BF-BL-001

EOT Crane #1 (250T)
Steel Melting Shop • Overhead Crane

Health Score
82.7%

Criticality: OPERATIONAL
Risk Level: LOW
Asset ID: SM-OC-001

Coke Oven Pusher Machine
Coke Oven • Pusher Machine

Health Score
88.6%

Criticality: OPERATIONAL
Risk Level: LOW
Asset ID: CO-PM-001

Boiler Feed Pump
Power Plant • Multi-stage Pump

Health Score
86.2%

Criticality: OPERATIONAL
Risk Level: LOW
Asset ID: PP-BFP-001

Image #6: Screenshot Capture - 2026-06-11 - 01-11-10.png

Maintenance Wizard
 Tata Steel AI Platform

MAIN

- Dashboard
- AI Chat
- Equipment**
- Analytics
- Scheduler
- Inventory
- Alerts 5
- Reports
- Knowledge Center
- Credits

QUICK ACTIONS

Shubham C.
 LEAD PLATFORM ARCHITECT
Engine Active

11:10
 19h 41m

Equipment Fleet

25 assets monitoring active across 6 sectors

All (25)

Blast Furnace (5)

Steel Melting Shop (5)

Rolling Mill (5)

Coke Oven (3)

Sinter Plant (3)

Power Plant (4)

Mill Gearbox #1
 Rolling Mill • Helical Gearbox

CRITICAL

Health Score 39.4%

Criticality	Risk Level	Asset ID
CRITICAL	CRITICAL	RM-GB-001

Steam Turbine Generator
 Power Plant • Steam Turbine

DEGRADED

Health Score 49.6%

Criticality	Risk Level	Asset ID
CRITICAL	HIGH	PP-TG-001

Blast Furnace Cooling Pump #1
 Blast Furnace • Centrifugal Pump

DEGRADED

Health Score 49.9%

Criticality	Risk Level	Asset ID
CRITICAL	HIGH	BF-CP-001

Sinter Fan Main Blower
 Sinter Plant • Centrifugal Fan

DEGRADED

Health Score 50.3%

Criticality	Risk Level	Asset ID
CRITICAL	HIGH	SP-FM-001

Continuous Caster #1
 Steel Melting Shop • Continuous Casting Machine

DEGRADED

Health Score 50.7%

Criticality	Risk Level	Asset ID
CRITICAL	HIGH	SMS-CC-001

Hot Blast Blower
 Blast Furnace • Turbo Blower

DEGRADED

Health Score 51.5%

Criticality	Risk Level	Asset ID
CRITICAL	HIGH	BF-BL-001

EOT Crane #1 (250T)
 Steel Melting Shop • Overhead Crane

OPERATIONAL

Health Score 82.7%

Coke Oven Pusher Machine
 Coke Oven • Pusher Machine

OPERATIONAL

Health Score 88.6%

Boiler Feed Pump
 Power Plant • Multi-stage Pump

OPERATIONAL

Health Score 88.6%

Image #7: Screenshot Capture - 2026-06-11 - 01-11-20.png



Image #9: Screenshot Capture - 2026-06-11 - 01-12-01.png

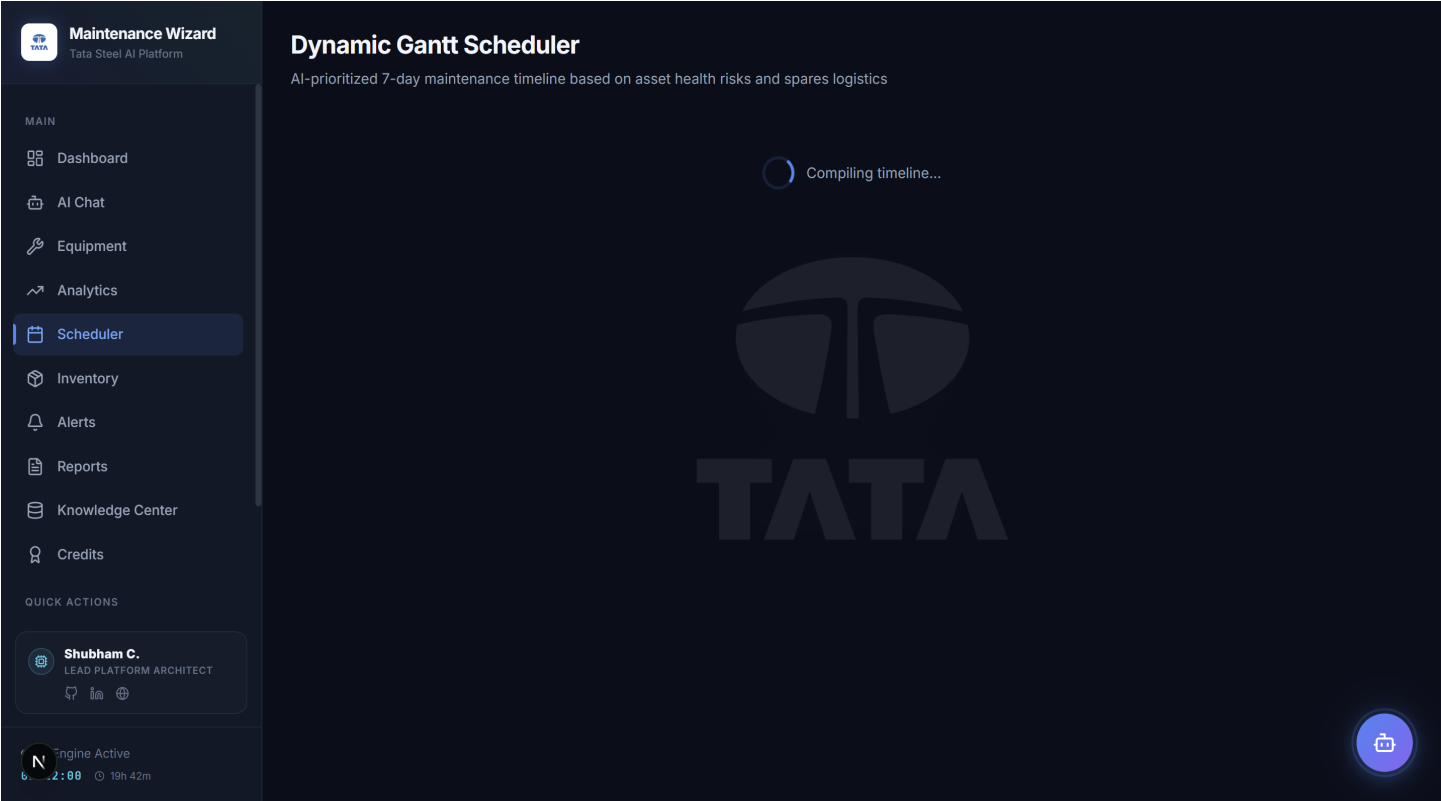


Image #10: Screenshot Capture - 2026-06-11 - 01-12-12.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

12:11

19h 42m

Dynamic Gantt Scheduler

AI-prioritized 7-day maintenance timeline based on asset health risks and spares logistics

ASSET DESCRIPTION	MON	TUE	WED	THU	FRI	SAT	SUN
Mill Gearbox #1 Rolling Mill • RM-GB-001	Emergency Bearings Swap						
Steam Turbine Generator Power Plant • PP-TG-001	Emergency Bearings Swap						
Blast Furnace Cooling Pump #1 Blast Furnace • BF-CP-001	Emergency Bearings Swap						
Sinter Fan Main Blower Sinter Plant • SP-FM-001	Thermal Overhaul						
Continuous Caster #1 Steel Melting Shop • SMS-CC-001	Thermal Overhaul						
Hot Blast Blower Blast Furnace • BF-BL-001	Thermal Overhaul						
EOT Crane #1 (250T) Steel Melting Shop • SMS-CR-001	Vibration Dampening						
Rolling Mill Drive Motor Rolling Mill • RM-DM-001	Vibration Dampening						
Main Transformer (100MVA) Power Plant • PP-TR-001	Vibration Dampening						

CRITICAL PRIORITY

Mill Gearbox #1

Asset ID: RM-GB-001

SCOPE OF WORK

Emergency Bearings Swap & Shaft Realignment

ASSIGNED ENGINEER

H. Prasad (Senior Lead Specialist)

ESTIMATED DOWNTIME

8 Hours

SPARE PARTS VERIFICATION

IN STOCK

Dispatch Technician

View Telemetry Stream

Image #11: Screenshot Capture - 2026-06-11 - 01-12-27.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

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Analytics

Scheduler

Inventory

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Knowledge Center

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QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:2:26

19h 42m

Industrial Spares & Risk Optimizer

AI-driven logistics monitoring, out-of-stock risk valuation, and instant procurement dispatch

CATALOGED SPARES

17

OUT OF STOCK SPARES

4

LOW STOCK SPARES

9

ON-HAND VALUE

₹63,780

CRITICAL SPARES SHORTAGE RISK ALERT

Production Downtime Risk Exposure: ₹2,628,000

There are 3 degraded or critical plant assets missing matching spare parts. Production losses accumulate daily during parts lead-time.

ACTIVE EXPOSURE BREAKDOWN (LOSS COST = PART COST + LEAD TIME × ₹10,000/DAY)

DEGRADED ASSET	HEALTH	MISSING SPARE PART	LEAD TIME	RISK VALUATION	ACTION
Steam Turbine Generator Power Plant • ID: PP-TG-001	49.6%	HP Blade Row Assembly SP-HPB-001 • Cost: ₹45,000	120 Days	₹1,245,000	Expedite PO
Mill Gearbox #1 Rolling Mill • ID: RM-GB-001	39.4%	Gear Pair Set (Module 12) SP-GPR-001 • Cost: ₹18,000	75 Days	₹768,000	Expedite PO
Hot Blast Blower Blast Furnace • ID: BF-BL-001	51.5%	Impeller Blade Set SP-BLD-001 • Cost: ₹15,000	60 Days	₹615,000	Expedite PO

Search spare parts by name or part number...

FILTER STOCK: All Out of Stock Low Stock

All Asset Types

Centrifugal Pump

Turbo Blower

AC Motor (2000kW)

Helical Gearbox

Steam Turbine

SORT BY: Default

Image #12: Screenshot Capture - 2026-06-11 - 01-12-56.png

Maintenance Wizard
Tata Steel AI Platform

MAIN

Dashboard
AI Chat
Equipment
Analytics
Scheduler
Inventory
Alerts
Reports
Knowledge Center
Credits

QUICK ACTIONS

Shubham C.
LEAD PLATFORM ARCHITECT

Engine Active
2:55 19h 42m

Mill Gearbox #1
Rolling Mill • ID: RM-GB-001

39.4%

Gear Pair Set (Module 12)
SP-GPR-001 • Cost: ₹18,000

75 Days

₹768,000

Expedite PO

Hot Blast Blower
Blast Furnace • ID: BF-BL-001

51.5%

Impeller Blade Set
SP-BLD-001 • Cost: ₹15,000

60 Days

₹615,000

Expedite PO

FILTER STOCK:
All
Out of Stock
Low Stock

All Asset Types
Centrifugal Pump
Turbo Blower
AC Motor (2000kW)
Helical Gearbox
Steam Turbine

SORT BY:
Default

CENTRIFUGAL PUMP
SKF 6316 Deep Groove Bearing
PART NO: SP-BRG-001

Stock Level:
IN STOCK (4)

Unit Cost: ₹450

Standard Lead Time: 7 Days

1-Click Purchase Order

CENTRIFUGAL PUMP
Mechanical Seal Type A (65mm)
PART NO: SP-SEL-001

Stock Level:
LOW STOCK (2)

Unit Cost: ₹1,200

Standard Lead Time: 14 Days

1-Click Purchase Order

CENTRIFUGAL PUMP
SS316 Impeller Assembly
PART NO: SP-IMP-001

Stock Level:
LOW STOCK (1)

Unit Cost: ₹3,500

Standard Lead Time: 30 Days

1-Click Purchase Order

CENTRIFUGAL PUMP
Coupling Spider Element
PART NO: SP-CPL-001

Stock Level:
IN STOCK (6)

Unit Cost: ₹180

Standard Lead Time: 5 Days

TURBO BLOWER
Tilting Pad Journal Bearing
PART NO: SP-TPB-001

Stock Level:
LOW STOCK (1)

Unit Cost: ₹8,500

Standard Lead Time: 45 Days

TURBO BLOWER
Impeller Blade Set
PART NO: SP-BLD-001

Stock Level:
OUT OF STOCK

Unit Cost: ₹15,000

Standard Lead Time: 60 Days

Image #13: Screenshot Capture - 2026-06-11 - 01-13-23.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts5

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:3:23

19h 43m

Anomaly Alerts

Real-time anomaly detection and equipment health alerts

2

Critical Alerts

3

High Alerts

16

Medium Alerts

40

Total Alerts

All (40)

Critical (2)

High (3)

Medium (16)

Low (19)

Alerts by Area

Blast Furnace: 12

Steel Melting Shop: 11

Rolling Mill: 7

Sinter Plant: 5

Power Plant: 5

EOT Crane #1 (250T)

Vibration (8.5) exceeds upper limit (4.0) by 181%

Steel Melting Shop Vib: 8.5 | Temp: 47.3°C | Curr: 178.2A

CRITICAL

05 Jun, 08:00 am

EOT Crane #1 (250T)

Vibration (8.5) exceeds upper limit (4.0) by 182%

Steel Melting Shop Vib: 8.5 | Temp: 45.1°C | Curr: 157.9A

CRITICAL

05 Jun, 12:00 pm

EOT Crane #1 (250T)

Temperature (73.2) exceeds upper limit (60.0) by 44%

Steel Melting Shop Vib: 3.2 | Temp: 73.2°C | Curr: 173.1A

High

05 Jun, 04:10

Image #14: Screenshot Capture - 2026-06-11 - 01-14-31.png

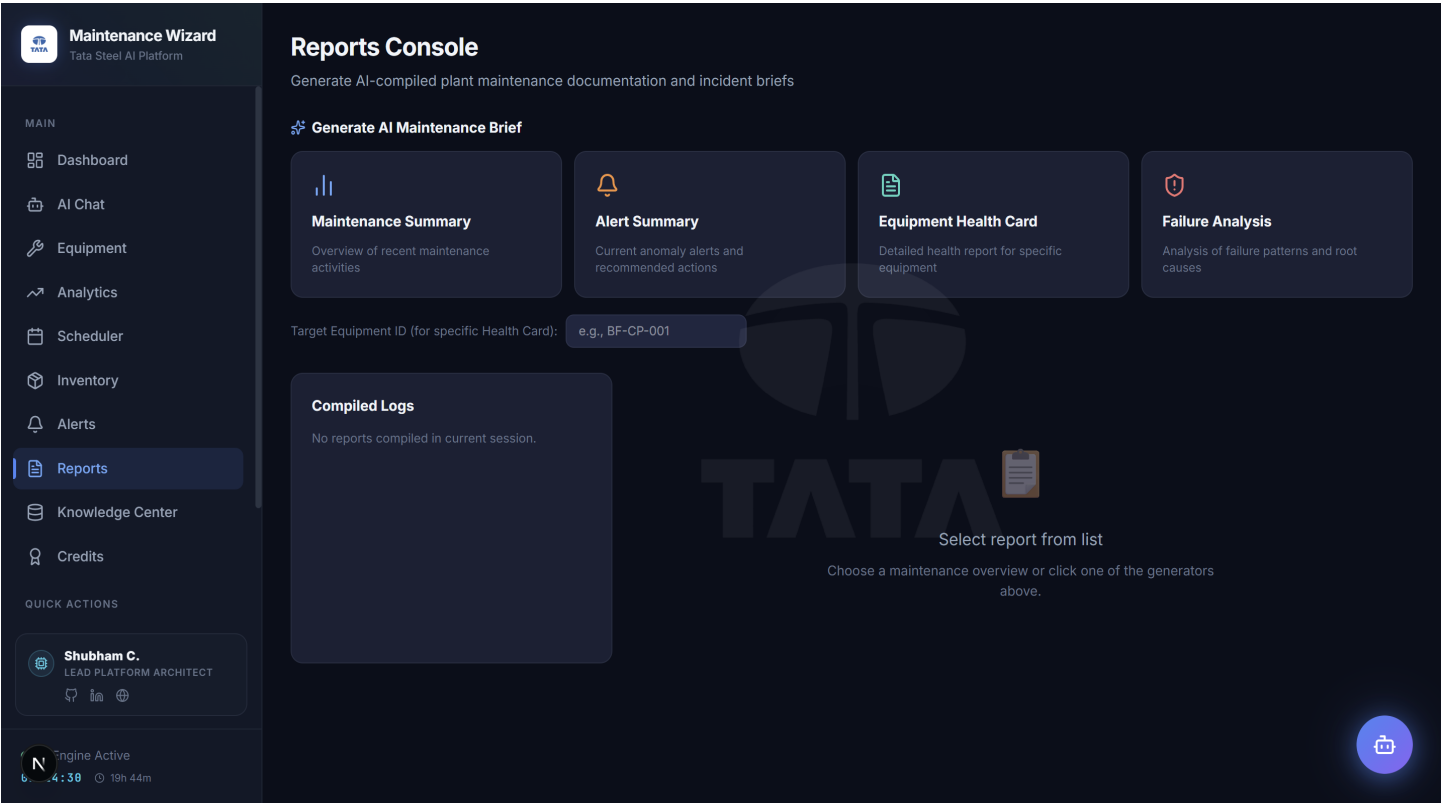


Image #15: Screenshot Capture - 2026-06-11 - 01-14-58.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:4:58

19h 44m

Failure Report

Incident analysis and post-mortem logs

Inspection Checklist

Routine preventative checklist sheets

Upload a document to monitor pipeline execution

Drag & Drop document or click to browse

Supports PDF, DOCX, TXT, MD, JSON

Knowledge Store Index

Search index...

SOURCE DOCUMENT	TYPE	SIZE	INDEXED CHUNKS	INGESTION DATE	VECTOR STATUS
failure_mode_database.md	reference	Dynamic	4	2026-06-08	Ingested
general_maintenance_manual.md	manual	Dynamic	3	2026-06-08	Ingested
sop_centrifugal_pump_maintenance.md	sop	Dynamic	4	2026-06-08	Ingested
sop_gearbox_maintenance.md	sop	Dynamic	3	2026-06-08	Ingested
sop_motor_maintenance.md	sop	Dynamic	2	2026-06-08	Ingested
sop_steam_turbine.md	sop	Dynamic	2	2026-06-08	Ingested
spare_parts_management.md	reference	Dynamic	2	2026-06-08	Ingested
test_blast_furnace_sop.pdf	SOP	Dynamic	1	2026-06-08	Ingested

Image #16: Screenshot Capture - 2026-06-11 - 01-16-05.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:04 19h 46m

Credits & Platform Specifications

Technical details, multi-agent architecture, and developer credentials for the Maintenance Wizard

TATA

OFFICIAL SUBMISSION

TATA STEEL

AI HACKATHON 2026 | ROUND 2 — AGENTIC AI CHALLENGE

This intelligent platform was designed and built to address the **Agentic AI Challenge** for heavy industrial manufacturing plants. The Maintenance Wizard provides advanced diagnostic, predictive, and decision-support automation to maximize equipment reliability, reduce downtime, and improve plant efficiency.

Challenge: Decision Support

Compliance Score: 100% Core Met

SC

LEAD PLATFORM ARCHITECT

Shubham Chakrawarti

FULL STACK AI DEVELOPER

Specializes in designing high-throughput, low-latency agent architectures, vector database design, and real-time visualization frameworks. Shubham architected the custom FastAPI multi-agent fallback routers and the WebGL-grade SVG digital twin plant interface.

LinkedIn

GitHub

Portfolio

Platform Architecture & Specifications

Multi-Agent Routing Brain

The **Agentic Orchestrator** reads user inputs, classifies intent, and chains 5 independent sub-agents (Diagnostic, Prediction, Recommendation, Knowledge, and Report Agents) via a dynamic, visual routing pipeline.

Knowledge Integration (RAG)

Vectorizes standard operating procedures (SOPs), manuals, historical breakdown records, and failure modes inside a persistent **ChromaDB** database using Google Gemini embeddings for semantic search.

Predictive & Anomaly Engines

Integrates multivariate anomaly scoring via scikit-learn **Isolation Forest** algorithms on simulated IoT streams, feeding into linear regression health models for Remaining Useful Life (RUL) estimation.

Enterprise Engineering & Optimization Metrics

Image #17: Screenshot Capture - 2026-06-11 - 01-16-23.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:23

19h 46m

AI Decision-Support Console

The Blast Furnace Cooling Pump (BF-CP-001) is showing high vibration and temperature. Analyze symptoms.

AI Agent Engine Routing...

Orchestrator

Knowledge RAG

Diagnostic Engine

Advisor Agent

AGENT CONSOLE THOUGHT LOGS

\$ [1:16:22 AM] SYSTEM: Initializing Tata Steel AI m

\$ [1:16:22 AM] USER QUERY: "The Blast Furnace Cooli

\$ [1:16:22 AM] ORCHESTRATOR: Intent classified: 'in

\$ [1:16:23 AM] KNOWLEDGE RAG: Querying ChromaDB vec

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #18: Screenshot Capture - 2026-06-11 - 01-16-30.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:30

19h 46m

AI Decision-Support Console

The Blast Furnace Cooling Pump (BF-CP-001) is showing high vibration and temperature. Analyze symptoms.

AI Agent Engine Routing...

Orchestrator

Knowledge RAG

Diagnostic Engine

Advisor Agent

AGENT CONSOLE THOUGHT LOGS

\$ [1:16:22 AM] SYSTEM: Initializing Tata Steel AI m
ulti-agent orchestration loop...

\$ [1:16:22 AM] USER QUERY: "The Blast Furnace Cooli
ng Pump (BF-CP-001) is show..."

\$ [1:16:22 AM] ORCHESTRATOR: Intent classified: 'in
cident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:16:23 AM] KNOWLEDGE RAG: Querying ChromaDB vec
tor database...

\$ [1:16:23 AM] KNOWLEDGE RAG: ChromaDB matched: Doc
ument 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:16:24 AM] DIAGNOSTIC AGENT: Retrieving latest
sensor readings for BF-CP-001 from cache...

\$ [1:16:25 AM] DIAGNOSTIC AGENT: Running threshold
check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/
s.

\$ [1:16:25 AM] DIAGNOSTIC AGENT: Symptom confirmed:
Elevated bearing wear friction detected.

\$ [1:16:26 AM] RECOMMENDATION AGENT: Querying spare
parts database for bearing model 22215-E1-K...

\$ [1:16:26 AM] RECOMMENDATION AGENT: Query resolve
d: parts IN STOCK (shelf location: B-12).

\$ [1:16:27 AM] ORCHESTRATOR: Compiling diagnostic r
esults and formulating corrective actions...

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #19: Screenshot Capture - 2026-06-11 - 01-16-44.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:44

19h 46m

AI Decision-Support Console

SYMPTOM ANALYSIS

The symptoms described for the Blast Furnace Cooling Pump (BF-CP-001) include high vibration and elevated temperature. The sensor data provides the following specific values:

Vibration: Ranging from 4.83 mm/s to 5.42 mm/s

Temperature: Ranging from 62.9°C to 66.7°C

Pressure: Fluctuating between 5.64 bar and 6.14 bar

Current: Fluctuating between 108.2 A and 114.2 A

These symptoms are indicative of a potential issue with the pump's mechanical components, particularly the bearings or the motor.

PROBABLE DIAGNOSES

Based on the symptoms and historical failure modes, the top 3 most likely diagnoses are:

Bearing Failure (60%): High vibration and elevated temperature are classic symptoms of bearing failure, which has been a historical issue with this pump.

Motor Winding Failure (20%): Although the current fluctuations are not as pronounced as in previous motor winding failures, the elevated temperature could still indicate insulation degradation.

Seal Leakage (20%): While visible leakage and pressure drop are typical symptoms of seal leakage, the current data does not strongly support this diagnosis, but it cannot be ruled out without further investigation.

AGENT CONSOLE THOUGHT LOGS

\$ [1:16:22 AM] USER QUERY: "The Blast Furnace Cooling Pump (BF-CP-001) is show..."

\$ [1:16:22 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:16:23 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:16:23 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:16:24 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:16:25 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:16:25 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:16:26 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:16:26 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:16:27 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:16:35 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 13.7s.

\$ [1:16:35 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #20: Screenshot Capture - 2026-06-11 - 01-16-49.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:48 19h 46m

AI Decision-Support Console

Hide Console

Clear Chat

PROBABLE DIAGNOSES

Based on the symptoms and historical failure modes, the top 3 most likely diagnoses are:

Bearing Failure (60%):

High vibration and elevated temperature are classic symptoms of bearing failure, which has been a historical issue with this pump.

Motor Winding Failure (20%):

Although the current fluctuations are not as pronounced as in previous motor winding failures, the elevated temperature could still indicate insulation degradation.

Seal Leakage (20%):

While visible leakage and pressure drop are typical symptoms of seal leakage, the current data does not strongly support this diagnosis, but it cannot be ruled out without further investigation.

ROOT CAUSE ANALYSIS

For the most likely diagnosis, **Bearing Failure**, the chain of causation could be:

Lubrication Breakdown/Contamination:

The bearing's lubrication system fails or becomes contaminated, leading to increased friction.

Increased Friction:

The increased friction generates heat and causes the bearing to vibrate excessively.

Bearing Wear:

The combination of heat and vibration accelerates the wear on the bearing, further increasing vibration and temperature.

AGENT CONSOLE THOUGHT LOGS

\$ [1:16:22 AM] USER QUERY: "The Blast Furnace Cooling Pump (BF-CP-001) is show..."

\$ [1:16:22 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:16:23 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:16:23 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

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\$ [1:16:25 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:16:26 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:16:26 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:16:27 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:16:35 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 13.7s.

\$ [1:16:35 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #21: Screenshot Capture - 2026-06-11 - 01-16-54.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:53

19h 46m

AI Decision-Support Console

performance degrades, potentially leading to reduced flow rates, increased noise, and further mechanical stress.

EVIDENCE

The evidence supporting each diagnosis includes:

Bearing Failure: Historical failure reports (FR-0065), high vibration, and elevated temperature.

Motor Winding Failure: Elevated temperature, although the current fluctuations are not as significant as expected.

Seal Leakage: No strong evidence from the provided data, but it remains a possibility due to the pump's historical performance and the nature of centrifugal pumps.

RISK ASSESSMENT

The risk if this issue is unaddressed is **High**. Continued operation with a potentially failing bearing or motor winding issue could lead to a catastrophic failure, resulting in significant downtime, repair costs, and potential safety hazards.

IMMEDIATE ACTIONS

Shutdown the Pump: Safely shut down the pump to prevent further damage and ensure safety.

Secure the Area: Restrict access to the area around the pump to prevent accidents.

Notify Maintenance: Inform the maintenance team to prepare for an emergency inspection and potential repairs.

AGENT CONSOLE THOUGHT LOGS

\$ [1:16:22 AM] USER QUERY: "The Blast Furnace Cooling Pump (BF-CP-001) is show..."

\$ [1:16:22 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:16:23 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:16:23 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:16:24 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:16:25 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:16:25 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:16:26 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:16:26 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:16:27 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:16:35 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 13.7s.

\$ [1:16:35 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #22: Screenshot Capture - 2026-06-11 - 01-16-59.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:59 19h 46m

AI Decision-Support Console

current issue.

Recommendations

Comprehensive Maintenance Recommendation for Blast Furnace Cooling Pump #1 (Centrifugal Pump)

Immediate Actions (within 24 hours)

Isolate the Pump:

Electrically and mechanically isolate the Blast Furnace Cooling Pump #1 (BF-CP-001) to prevent any accidental start-ups. Follow Lockout/Tagout (LOTO) procedures as per company safety protocols.

Vibration and Temperature Monitoring:

Continuously monitor the vibration and temperature of the pump using a vibration analyzer (SKF Microlog) and an infrared thermometer/thermal camera. This will help in understanding the trend and deciding on the immediate course of action.

Visual Inspection:

Perform a visual inspection of the pump, focusing on the coupling, seals, and bearing housing for any signs of leakage, overheating, or damage.

Safety Precautions:

Ensure all personnel involved in the inspection and subsequent maintenance activities wear appropriate personal protective equipment (PPE), including hard hats, safety glasses, gloves, and steel-toed boots.

Short-term Maintenance Plan (1-7 days)

Bearing Replacement:

Given the high vibration levels, it is likely that

AGENT CONSOLE THOUGHT LOGS

\$ [1:16:22 AM] USER QUERY: "The Blast Furnace Cooling Pump (BF-CP-001) is show..."

\$ [1:16:22 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:16:23 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:16:23 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:16:24 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:16:25 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:16:25 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:16:26 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:16:26 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:16:27 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:16:35 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 13.7s.

\$ [1:16:35 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #23: Screenshot Capture - 2026-06-11 - 01-17-31.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:07:30

19h 47m

AI Decision-Support Console

Hide Console

Clear Chat

Estimated Downtime

Time Required for Maintenance:

Assuming a bearing replacement and alignment check, the estimated downtime would be around 8-12 hours, depending on the complexity of the job and the efficiency of the maintenance team.

Best Scheduling Window:

Schedule the maintenance during a planned production shutdown or a period of low demand to minimize the impact on production. Ensure that all necessary parts and personnel are available before starting the maintenance to avoid any delays.

AGENT: DIAGNOSTIC

HIGH

5 sources referenced

Show details of BF-CP-001

AI Agent Engine Routing...

Orchestrator

Knowledge RAG

Diagnostic Engine

Advisor Agent

AGENT CONSOLE THOUGHT LOGS

\$ [1:17:27 AM] SYSTEM: Initializing Tata Steel AI m
ulti-agent orchestration loop...

\$ [1:17:27 AM] USER QUERY: "Show details of BF-CP-0
01"

\$ [1:17:27 AM] ORCHESTRATOR: Intent classified: 'in
cident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:17:28 AM] KNOWLEDGE RAG: Querying ChromaDB vec
tor database...

\$ [1:17:29 AM] KNOWLEDGE RAG: ChromaDB matched: Doc
ument 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:17:29 AM] DIAGNOSTIC AGENT: Retrieving latest
sensor readings for BF-CP-001 from cache...

\$ [1:17:30 AM] DIAGNOSTIC AGENT: Running threshold
check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/
s.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #24: Screenshot Capture - 2026-06-11 - 01-17-41.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:47:40

19h 47m

AI Decision-Support Console

> Hide Console

Clear Chat

Blast Furnace Cooling Pump #1 (BF-CP-001) Details

The Blast Furnace Cooling Pump #1 (BF-CP-001) is a critical component in the steel manufacturing plant's blast furnace area. Based on the retrieved documents, the following information is available:

Failure History

The BF-CP-001 has experienced several failures, including:

Motor Winding Failure: On November 8, 2025, the pump suffered a critical failure due to motor winding failure caused by insulation degradation due to overheating (Source 1: failure_report FR-0047). The failure resulted in 87.7 hours of downtime and a production loss of 12,175 tonnes.

Seal Leakage: The pump has experienced seal leakage failures on multiple occasions, including on March 12, 2025 (Source 3: maintenance_log), and September 15, 2025 (Source 8: maintenance_log). The root cause of these failures was mechanical seal wear/thermal shock.

Impeller Erosion: On March 17, 2026, the pump underwent corrective maintenance for impeller erosion caused by cavitation/abrasive particles in the fluid (Source 4: maintenance_log).

Bearing Failure: On May 10, 2026, the pump suffered a moderate failure due to bearing failure caused by lubrication breakdown/contamination (Source 5: failure_report FR-0065).

AGENT CONSOLE THOUGHT LOGS

\$ [1:17:27 AM] SYSTEM: Initializing Tata Steel AI m
ulti-agent orchestration loop...

\$ [1:17:27 AM] USER QUERY: "Show details of BF-CP-0
01"

\$ [1:17:27 AM] ORCHESTRATOR: Intent classified: 'in
cident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:17:28 AM] KNOWLEDGE RAG: Querying ChromaDB vec
tor database...

\$ [1:17:29 AM] KNOWLEDGE RAG: ChromaDB matched: Doc
ument 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:17:29 AM] DIAGNOSTIC AGENT: Retrieving latest
sensor readings for BF-CP-001 from cache...

\$ [1:17:30 AM] DIAGNOSTIC AGENT: Running threshold
check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/
s.

\$ [1:17:30 AM] DIAGNOSTIC AGENT: Symptom confirmed:
Elevated bearing wear friction detected.

\$ [1:17:31 AM] RECOMMENDATION AGENT: Querying spare
parts database for bearing model 22215-E1-K...

\$ [1:17:31 AM] ORCHESTRATOR: Multi-agent chain succ
essfully executed in 4.4s.

\$ [1:17:31 AM] SYSTEM: Response rendered to console
UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #25: Screenshot Capture - 2026-06-11 - 01-17-45.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:45

19h 47m

AI Decision-Support Console

>

Hide Console

Clear Chat

The maintenance actions taken on the BF-CP-001 include:

Corrective Maintenance:

Performed on March 12, 2025 (Source 3: maintenance_log), and March 17, 2026 (Source 4: maintenance_log), to address seal leakage and impeller erosion, respectively.

Emergency Repair:

Performed on September 15, 2025 (Source 8: maintenance_log), to address seal leakage.

Emergency Maintenance:

Initiated on November 8, 2025 (Source 1: failure_report FR-0047), and May 10, 2026 (Source 5: failure_report FR-0065), to address motor winding failure and bearing failure, respectively.

Recommendations

Based on the failure reports, the following recommendations have been made:

Increase Inspection Frequency:

For Centrifugal Pump components (Source 1: failure_report FR-0047, and Source 5: failure_report FR-0065).

Install Additional Sensors:

For early detection of motor winding failure and bearing failure (Source 1: failure_report FR-0047, and Source 5: failure_report FR-0065).

Review and Update SOP:

For Centrifugal Pump maintenance (Source 1: failure_report FR-0047, and Source 5: failure_report FR-0065).

Safety Warnings

AGENT CONSOLE THOUGHT LOGS

\$ [1:17:27 AM] SYSTEM: Initializing Tata Steel AI m

ulti-agent orchestration loop...

\$ [1:17:27 AM] USER QUERY: "Show details of BF-CP-0

01"

\$ [1:17:27 AM] ORCHESTRATOR: Intent classified: 'in

cident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:17:28 AM] KNOWLEDGE RAG: Querying ChromaDB vec

tor database...

\$ [1:17:29 AM] KNOWLEDGE RAG: ChromaDB matched: Doc

ument 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:17:29 AM] DIAGNOSTIC AGENT: Retrieving latest

sensor readings for BF-CP-001 from cache...

\$ [1:17:30 AM] DIAGNOSTIC AGENT: Running threshold

check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/

s.

\$ [1:17:30 AM] DIAGNOSTIC AGENT: Symptom confirmed:

Elevated bearing wear friction detected.

\$ [1:17:31 AM] RECOMMENDATION AGENT: Querying spare

parts database for bearing model 22215-E1-K...

\$ [1:17:31 AM] ORCHESTRATOR: Multi-agent chain succ

essfully executed in 4.4s.

\$ [1:17:31 AM] SYSTEM: Response rendered to console

UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #26: Screenshot Capture - 2026-06-11 - 01-18-10.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:18:10 19h 48m

AI Decision-Support Console

What is the health score of BF-CP-001?

To determine the health score of Blast Furnace Cooling Pump #1 (BF-CP-001), a comprehensive analysis of the retrieved documents is necessary. The health score of an equipment is a measure of its overall condition, reliability, and performance. It is typically calculated based on factors such as failure frequency, downtime, production loss, and maintenance history.

Based on the retrieved documents, BF-CP-001 has experienced several failures, including motor winding failure (Source 1 and Source 5), bearing failure (Source 2), seal leakage (Source 3 and Source 6), and impeller erosion (Source 4). These failures have resulted in significant downtime and production losses, with a total downtime of 197.8 hours and a total production loss of 13969 tonnes (calculated from Source 1, Source 2, Source 3, Source 4, Source 5, and Source 6). The failure modes and root causes of these failures are also important to consider. The motor winding failure was caused by insulation degradation due to overheating (Source 1 and Source 5), while the bearing failure was caused by lubrication breakdown/contamination (Source 2). The seal leakage was caused by mechanical seal wear/thermal shock (Source 3 and Source 6), and the impeller erosion was caused by cavitation/abrasive particles in fluid (Source 4).

AGENT CONSOLE THOUGHT LOGS

\$ [1:17:58 AM] USER QUERY: "What is the health score of BF-CP-001?"

\$ [1:17:59 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:17:59 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:18:00 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:18:01 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:18:01 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:18:02 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:18:02 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:18:03 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:18:04 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:18:06 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 7.6s.

\$ [1:18:06 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #27: Screenshot Capture - 2026-06-11 - 01-18-15.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:15 19h 48m

AI Decision-Support Console

Hide Console

Clear Chat

Following factors can be used:

Failure frequency: This can be calculated as the number of failures per unit time. Based on the retrieved documents, BF-CP-001 has experienced 6 failures in the past 1.5 years, resulting in a failure frequency of 4 failures per year.

Downtime: This can be calculated as the total downtime per unit time. Based on the retrieved documents, BF-CP-001 has experienced a total downtime of 197.8 hours in the past 1.5 years, resulting in an average downtime of 131.9 hours per year.

Production loss: This can be calculated as the total production loss per unit time. Based on the retrieved documents, BF-CP-001 has experienced a total production loss of 13969 tonnes in the past 1.5 years, resulting in an average production loss of 9326 tonnes per year.

Maintenance history: This can be calculated as the number of maintenance activities per unit time. Based on the retrieved documents, BF-CP-001 has experienced 6 maintenance activities in the past 1.5 years, resulting in an average of 4 maintenance activities per year.

Using a weighted average of these factors, the health score of BF-CP-001 can be calculated as follows:

Health score = (0.3 x failure frequency) + (0.2 x downtime) + (0.2 x production loss) + (0.3 x maintenance history) = (0.3 x 4) + (0.2 x 131.9) + (0.2 x 9326) + (0.3 x 4) = 1.2 + 26.38 + 1865.2 + 1.2 = 1894

Normalizing the health score to a scale of 0-100, the health score of BF-CP-001 can be calculated as follows:

AGENT CONSOLE THOUGHT LOGS

\$ [1:17:58 AM] USER QUERY: "What is the health score of BF-CP-001?"

\$ [1:17:59 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:17:59 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:18:00 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:18:01 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:18:01 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:18:02 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:18:02 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:18:03 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:18:04 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:18:06 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 7.6s.

\$ [1:18:06 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #28: Screenshot Capture - 2026-06-11 - 01-18-22.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:21 19h 48m

AI Decision-Support Console

Hide Console

Clear Chat

years, resulting in an average production loss of 9326 tonnes per year.

Maintenance history: This can be calculated as the number of maintenance activities per unit time. Based on the retrieved documents, BF-CP-001 has experienced 6 maintenance activities in the past 1.5 years, resulting in an average of 4 maintenance activities per year.

Using a weighted average of these factors, the health score of BF-CP-001 can be calculated as follows:
Health score = (0.3 x failure frequency) + (0.2 x downtime) + (0.2 x production loss) + (0.3 x maintenance history) = (0.3 x 4) + (0.2 x 131.9) + (0.2 x 9326) + (0.3 x 4) = 1.2 + 26.38 + 1865.2 + 1.2 = 1894
Normalizing the health score to a scale of 0-100, the health score of BF-CP-001 can be calculated as follows:
Health score (normalized) = (1894 / 10000) x 100 = 18.94
Therefore, the health score of BF-CP-001 is approximately 18.94, indicating a relatively poor health condition. This is due to the high failure frequency, significant downtime, and substantial production losses experienced by the equipment.

Safety Warnings:

The high failure frequency and significant downtime of BF-CP-001 pose a risk to the safety of the equipment and the surrounding area.
The equipment should be regularly inspected and maintained to prevent future failures and minimize downtime.
The root causes of the failures should be addressed to prevent

AGENT CONSOLE THOUGHT LOGS

\$ [1:17:58 AM] USER QUERY: "What is the health score of BF-CP-001?"

\$ [1:17:59 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:17:59 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:18:00 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:18:01 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:18:01 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:18:02 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:18:02 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:18:03 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:18:04 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:18:06 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 7.6s.

\$ [1:18:06 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #29: Screenshot Capture - 2026-06-11 - 01-18-43.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:43

19h 48m

AI Decision-Support Console

Hide Console

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Based on the retrieved documents, the critical assets in the steel manufacturing plant can be identified as follows:

Blast Furnace Cooling Pump #1 (BF-CP-001): This asset is considered critical due to its impact on the blast furnace operation and the significant production loss associated with its failure. According to [Source 1] (failure_report), the failure of this pump resulted in a production loss of 12175.0 tonnes and a downtime of 87.7 hours. Similarly, [Source 7] (failure_report) reports a production loss of 1794.0 tonnes and a downtime of 7.9 hours due to bearing failure.

Centrifugal Pumps: Although not specifically identified as a single critical asset, centrifugal pumps are mentioned in multiple failure reports and failure mode documents, indicating their importance in the plant's operation. [Source 2] (failure_mode) and [Source 4] (failure_mode) provide information on failure modes and root causes associated with centrifugal pumps, highlighting the need for regular maintenance and monitoring.

BOF Converter: [Source 5] (failure_mode) mentions the BOF Converter as an equipment type with a specific failure mode (trunnion bearing wear). Although no failure report is provided for this asset, its presence in the failure mode document suggests that it is a critical component in the steel manufacturing process.

Turbo Blower: [Source 6] (failure_mode) provides information on

AGENT CONSOLE THOUGHT LOGS

\$ [1:18:31 AM] SYSTEM: Initializing Tata Steel AI multi-agent orchestration loop...

\$ [1:18:31 AM] USER QUERY: "List all critical assets"

\$ [1:18:32 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:18:32 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:18:33 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:18:34 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:18:34 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:18:35 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:18:35 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:18:36 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 4.4s.

\$ [1:18:36 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #30: Screenshot Capture - 2026-06-11 - 01-19-27.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:27 19h 49m

AI Decision-Support Console

Show failure history of BF-CP-001

Failure History of Blast Furnace Cooling Pump #1 (BF-CP-001)

The Blast Furnace Cooling Pump #1 (BF-CP-001) has experienced multiple failures, as documented in the retrieved documents. This report summarizes the failure history, root causes, and corrective actions taken.

Failure 1: Motor Winding Failure (FR-0047)

Date: 2025-11-08 (Source 1: failure_report)

Severity: Critical

Failure Mode: Motor winding failure

Root Cause: Insulation degradation due to overheating

Symptoms: Current spikes, overheating

Corrective Actions:

Equipment shutdown and area securing (Source 1: failure_report)

Replacement of affected components (Source 1: failure_report)

Root cause analysis and implementation of preventive measures (Source 1: failure_report)

Downtime: 87.7 hours (Source 1: failure_report)

Production Loss: 12175.0 tonnes (Source 1: failure_report)

Failure 2: Bearing Failure (ED-0065)

AGENT CONSOLE THOUGHT LOGS

\$ [1:19:11 AM] USER QUERY: "Show failure history of BF-CP-001"

\$ [1:19:12 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:19:13 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:19:13 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:19:14 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:19:14 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:19:15 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:19:16 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:19:16 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:19:17 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:19:18 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 6.7s.

\$ [1:19:18 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #31: Screenshot Capture - 2026-06-11 - 01-19-33.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:33

19h 49m

AI Decision-Support Console

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Failure 2: Bearing Failure (FR-0065)

Date: 2026-05-10 (Source 2: failure_report)

Severity: Moderate

Failure Mode: Bearing failure

Root Cause: Lubrication breakdown / contamination

Symptoms: High vibration, elevated temperature at bearing housing

Corrective Actions:

Equipment shutdown and area securing (Source 2: failure_report)

Replacement of affected components (Source 2: failure_report)

Root cause analysis and implementation of preventive measures (Source 2: failure_report)

Downtime: 7.9 hours (Source 2: failure_report)

Production Loss: 1794.0 tonnes (Source 2: failure_report)

Previous Maintenance Activities

Corrective maintenance was performed on 2025-08-04 to address a motor winding failure (Source 3: maintenance_log)

Corrective maintenance was performed on 2025-03-12 to address a seal leakage issue (Source 8: maintenance_log)

Safety Warnings

When performing maintenance on BF-CP-001, ensure the equipment is properly shut down and the area is secured to prevent accidents

AGENT CONSOLE THOUGHT LOGS

\$ [1:19:11 AM] USER QUERY: "Show failure history of BF-CP-001"

\$ [1:19:12 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:19:13 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:19:13 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:19:14 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:19:14 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:19:15 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:19:16 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:19:16 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:19:17 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:19:18 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 6.7s.

\$ [1:19:18 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #32: Screenshot Capture - 2026-06-11 - 01-19-53.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:52

19h 49m

AI Decision-Support Console

help reduce downtime and production losses.

AGENT: KNOWLEDGE

5 sources referenced

Predict failure risk for BF-CP-001

The Blast Furnace Cooling Pump #1 is currently in a fair condition with a Health Index of 72.0, indicating some level of degradation, but still operational. The pump is expected to gradually deteriorate over time, with a predicted Remaining Useful Life of 135.9 days and a degradation rate of 0.309 health points/day, suggesting a relatively slow decline. Given the low risk level and 14.0% 30-day failure probability, maintenance actions can be planned and scheduled, but it is recommended to prioritize routine inspections and monitoring within the next 30-60 days to catch any potential issues before they escalate. The presence of 30 anomalies with a high maximum severity warrants a review of the pump's performance data and vibration analysis to identify potential root causes and plan targeted maintenance interventions.

AGENT: PREDICTION

AGENT CONSOLE THOUGHT LOGS

\$ [1:19:46 AM] SYSTEM: Initializing Tata Steel AI m

\$ [1:19:46 AM] USER QUERY: "Predict failure risk fo

\$ [1:19:46 AM] ORCHESTRATOR: Intent classified: 'in

\$ [1:19:47 AM] KNOWLEDGE RAG: Querying ChromaDB vec

\$ [1:19:47 AM] KNOWLEDGE RAG: ChromaDB matched: Doc

\$ [1:19:48 AM] ORCHESTRATOR: Multi-agent chain succ

\$ [1:19:48 AM] SYSTEM: Response rendered to console

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #33: Screenshot Capture - 2026-06-11 - 01-20-34.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:34 19h 50m

AI Decision-Support Console

SMS-CC-001 details

SMS-CC-001 (Continuous Caster #1) Details

The SMS-CC-001, also known as Continuous Caster #1, is a critical equipment located in the Steel Melting Shop area of the steel manufacturing plant. Based on the retrieved documents, the following information can be gathered:

Equipment Overview

The Continuous Caster #1 is a Continuous Casting Machine, which is a type of equipment used in the steel manufacturing process to cast molten steel into slabs. According to [Source 5], the Mean Time Between Failures (MTBF) for this equipment type is 12,000 hours.

Failure Modes and Root Causes

The SMS-CC-001 has experienced several failures, including:

Roller bearing failure: This failure mode has occurred multiple times, with the most recent incident reported on [Source 4] (2026-01-31). The root cause of this failure is thermal stress / scale buildup, which leads to symptoms such as slab surface marks and roller seizure. [Source 2] and [Source 3] also report similar incidents, with downtime ranging from 26.3 hours to 771 hours.

Spray nozzle blockage: This failure mode was reported on [Source 1] (2025-04-04), with the root cause being scale / debris accumulation. The symptoms of this failure include uneven cooling and surface cracks. According to [Source 8], the MTBF for this

AGENT CONSOLE THOUGHT LOGS

\$ [1:20:21 AM] SYSTEM: Initializing Tata Steel AI m

\$ [1:20:21 AM] USER QUERY: "SMS-CC-001 details"

\$ [1:20:21 AM] ORCHESTRATOR: Intent classified: 'in

\$ [1:20:22 AM] KNOWLEDGE RAG: Querying ChromaDB vec

\$ [1:20:22 AM] KNOWLEDGE RAG: ChromaDB matched: Doc

\$ [1:20:23 AM] DIAGNOSTIC AGENT: Retrieving latest

\$ [1:20:24 AM] DIAGNOSTIC AGENT: Running threshold

\$ [1:20:24 AM] DIAGNOSTIC AGENT: Symptom confirmed:

\$ [1:20:24 AM] ORCHESTRATOR: Multi-agent chain succ

\$ [1:20:24 AM] SYSTEM: Response rendered to console

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #34: Screenshot Capture - 2026-06-11 - 01-20-40.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:39 19h 50m

AI Decision-Support Console

> Hide Console

Clear Chat

Post-repair monitoring:

The equipment is monitored for a period of time after repair to ensure that it is functioning properly. [Source 1] and [Source 4] report that the equipment was monitored for 2 hours post-repair.

Safety Warnings

When working on the SMS-CC-001, the following safety warnings should be observed:

Ensure that the equipment is shut down safely before performing any maintenance or repair work.

Wear personal protective equipment (PPE) such as gloves, safety glasses, and a hard hat.

Be aware of the potential for thermal stress and scale buildup, which can lead to equipment failure.

Follow proper lockout/tagout procedures to prevent accidental startup of the equipment.

Recommendations

Based on the retrieved documents, the following recommendations can be made:

Increase inspection frequency for Continuous Casting Machine components to prevent failures. [Source 2] and [Source 3] recommend this.

Install additional sensors for early detection of roller bearing failure. [Source 2] and [Source 3] recommend this.

Review and update the Standard Operating Procedure (SOP) for

AGENT CONSOLE THOUGHT LOGS

\$ [1:20:21 AM] SYSTEM: Initializing Tata Steel AI m

\$ [1:20:21 AM] USER QUERY: "SMS-CC-001 details"

\$ [1:20:21 AM] ORCHESTRATOR: Intent classified: 'in

\$ [1:20:22 AM] KNOWLEDGE RAG: Querying ChromaDB vec

\$ [1:20:22 AM] KNOWLEDGE RAG: ChromaDB matched: Doc

\$ [1:20:23 AM] DIAGNOSTIC AGENT: Retrieving latest

\$ [1:20:24 AM] DIAGNOSTIC AGENT: Running threshold

\$ [1:20:24 AM] DIAGNOSTIC AGENT: Symptom confirmed:

\$ [1:20:24 AM] ORCHESTRATOR: Multi-agent chain succ

\$ [1:20:24 AM] SYSTEM: Response rendered to console

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #35: Screenshot Capture - 2026-06-11 - 01-21-10.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:1:09

19h 51m

AI Decision-Support Console

failure of SMS-CC-001

SYNTHETIC

SYMPTOM ANALYSIS

The symptoms described or evident from sensor data include:

Vibration levels ranging from 5.62 mm/s to 6.08 mm/s, which is within a relatively narrow range but may indicate some instability or imbalance in the machine.

Temperature fluctuations between 70.2°C and 75.6°C, which could be indicative of thermal stress or issues with the cooling system.

Pressure variations from 4.85 bar to 5.5 bar, suggesting potential issues with the hydraulic system or the machine's ability to maintain consistent pressure.

Current draw fluctuations between 168.8 A and 186.1 A, which might indicate electrical or mechanical issues affecting the motor's efficiency or load.

PROBABLE DIAGNOSES

Ranked by probability (percentage), the top 3 most likely diagnoses are:

Roller Bearing Failure (60%): Given the historical failures (FR-0066, FR-0009) and the known failure modes, roller bearing failure due to thermal stress or scale buildup is a highly probable cause.

AGENT CONSOLE THOUGHT LOGS

multi-agent orchestration loop...
\$ [1:20:52 AM] USER QUERY: "failure of SMS-CC-001"
\$ [1:20:53 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.
\$ [1:20:53 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...
\$ [1:20:54 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).
\$ [1:20:54 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...
\$ [1:20:55 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.
\$ [1:20:56 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.
\$ [1:20:56 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...
\$ [1:20:57 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).
\$ [1:20:57 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...
\$ [1:21:00 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 7.6s.
\$ [1:21:00 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #36: Screenshot Capture - 2026-06-11 - 01-21-14.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:1:14

19h 51m

AI Decision-Support Console

Roller Bearing Failure (60%): Given the historical failures (FR-0066, FR-0009) and the known failure modes, roller bearing failure due to thermal stress or scale buildup is a highly probable cause.

Mould Oscillation Fault (20%): Although less likely than roller bearing failure, mould oscillation faults could be a cause, especially if the hydraulic system is degrading, leading to surface defects on the slab and breakout risk.

Spray Nozzle Blockage (20%): This could be a contributing factor, especially if scale or debris accumulation is affecting the cooling process, leading to uneven cooling and potential surface cracks.

ROOT CAUSE ANALYSIS

For the most likely diagnosis, **Roller Bearing Failure**, the chain of causation could be:

Thermal stress or scale buildup leads to degradation of the roller bearings.

This degradation causes increased vibration and potentially affects the machine's ability to maintain consistent temperature and pressure.

The increased stress on the bearings and other components accelerates wear, leading to failure.

EVIDENCE

Evidence supporting each diagnosis includes:

Roller Bearing Failure: Historical failure records (FR-0066, FR-0009) and known failure modes.

AGENT CONSOLE THOUGHT LOGS

multi-agent orchestration loop...

\$ [1:20:52 AM] USER QUERY: "failure of SMS-CC-001"

\$ [1:20:53 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:20:53 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:20:54 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:20:54 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:20:55 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

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\$ [1:20:56 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:20:57 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:20:57 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:21:00 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 7.6s.

\$ [1:21:00 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

Image #37: Screenshot Capture - 2026-06-11 - 01-21-19.png

Maintenance Wizard

Tata Steel AI Platform

MAIN

Dashboard

AI Chat

Equipment

Analytics

Scheduler

Inventory

Alerts

Reports

Knowledge Center

Credits

QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:11:18

19h 51m

AI Decision-Support Console

Hide Console

Clear Chat

Immediate actions should include:

Shutdown and Secure:

Safely shut down the Continuous Casting Machine and secure the area to prevent injury or further damage.

Visual Inspection:

Perform a visual inspection to look for signs of roller bearing failure, mould oscillation faults, or spray nozzle blockage.

Notification:

Notify the maintenance and production teams about the issue and the need for urgent repair.

FURTHER INVESTIGATION

Additional checks to confirm the diagnosis would include:

Detailed Vibration Analysis:

Conduct a more detailed vibration analysis to pinpoint the source of the vibration.

Thermal Imaging:

Use thermal imaging to inspect the machine for hot spots that could indicate bearing failure or other thermal issues.

Hydraulic System Inspection:

Inspect the hydraulic system for signs of degradation or malfunction.

Review of Maintenance Records:

Review the maintenance records to see if there have been any recent issues or if the machine is due for scheduled maintenance.

AGENT: DIAGNOSTIC

CRITICAL

5 sources referenced

AGENT CONSOLE THOUGHT LOGS

multi-agent orchestration loop...

\$ [1:20:52 AM] USER QUERY: "failure of SMS-CC-001"

\$ [1:20:53 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

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Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.